TLX-8

High Volume Fiberglass Reinforced Microwave Substrate

AGC
Your Dreams, Our Challenge

Benefits

- Low and Stable Dk
- Excellent Mechanical & Thermal Properties
- Dimensionally Stable
- Low Moisture Absorption
- Low DF
- UL 94 V-0 Rating
- For Low Layer Count Microwave Designs

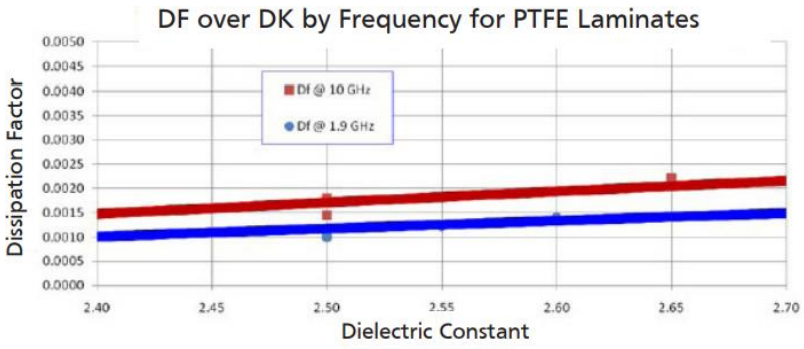
Applications

- Antennas
- Mixers, Splitters, Filters & Combiners
- Passive Components

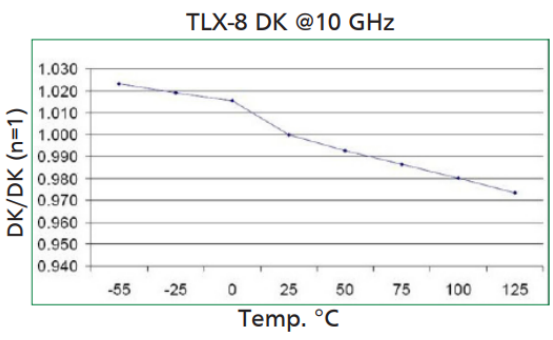
TLX-8 offers reliability in a wide range of RF applications. This material is versatile due to its 2.45 ~ 2.65 DK range and available thicknesses and copper cladding. It is suitable for low layer count microwave designs. TLX-8 PTFE fiberglass laminates are ideal for use in radar systems, mobile communications, microwave test equipment, microwave transmission devices and RF components. TLX-8 is a workhorse in the RF microwave substrate world where the fiberglass offers mechanical reinforcement. wherever a substrate experience severe environment such as:

- Resistance to creep for PWBs bolted to a housing that encounters high levels of vibration during space launch
- High temperature exposure in engine modules
- Radiation resistance in space (see NASA's website for low outgassing materials)
- Resistance to extreme environments at sea for warship antennas
- Resistance to a wide temperature range for altimeter substrates during flight.

The wide range of dielectric constants available enable the manufacture of couplers, splitters, combiners, amplifiers, antennas and other components.



TLX-8 has a long space heritage and is used wherever a woven fiberglass reinforcement is required.



TLX-8 has ± 2% variation in DK from -55 to 125 °C. If more temperature stability is required, TSM-DS3 should be considered.

Properties	Conditions	Typical Value	Unit	Test Method
Electrical Properties				
Dielectric Constant	@ 1MHz	2.55 ± 0.04		IPC-650 2.5.5.3
Dissipation Factor	@ 10 GHz	0.0018		IPC-650 2.5.5.5.1(Modified)
Outgassing	% TML	0.03	4 H 257 °F @ ≤ 5 x 10-5 Torr	ASTM E 595
	% CVCM	0.00		
	% WVR	0.01		
Surface Resistivity		6.605×10^{8}	Mohm	IPC-650 2.5.17.1 Sec. 5.2.1 (Elevated Temp.)
		3.550×10^{6}	Mohm	IPC-650 2.5.17.1 Sec. 5.2.1 (Humidity Cond.)
Volume Resistivity		1.110×10^{10}	Mohm-cm	IPC-650 2.5.17.1 Sec. 5.2.1 (Elevated Temp.)
		1.046×10^{10}	Mohm-cm	IPC-650 2.5.17.1 Sec. 5.2.1 (Humidity Cond.)
Dimensional Stability	MD	0.06	mm/M	IPC-650 2.4.39 Sec. 5.4 (After Bake)
	CD	0.08	mm/M	
	MD	0.09	mm/M	IPC-650 2.4.39 Sec. 5.5 (Thermal Stress)
	CD	0.10	mm/M	
Thermal Properties				
Thermal Conductivity	Unclad	0.19	W/M*K	ASTM F 433/ASTM 1530-06
CTE (25-260 °C)	X	21	ppm/°C	IPC-650 2.4 .41/TMA
	Y	23		
	Z	215		
T_d	2% Weight Loss	535	°C	IPC-650 2.4.24.6 (TGA)
	5% Weight Loss	553	°C	
Mechanical Properties				
Peel Strength	1 oz. ED	2.63 (15)	N/mm (lb/in)	IPC-650 2.4.8 Sec. 5.2.2 (Thermal Stress)
	1 oz. RTF	2.98 (17)	N/mm (lb/in)	
	½ oz. ED	2.45 (14)	N/mm (lb/in)	IPC-650 2.4.8.3 (Elevated Temp.)
	½ oz. ED	1.93 (11)	N/mm (lb/in)	IPC-650 2.4.8 Sec. 5.2.2 (Thermal Stress)
	1 oz. rolled	2.28 (13)	N/mm (lb/in)	
Young's Modulus	MD	6,757 (980)	N/mm^2 (psi)	ASTM D 902
	CD	8,274 (1,200)	N/mm^2 (psi)	
	MD	11,238 (1,630)	N/mm^2 (psi)	ASTM D 3039
Chemical / Physical Properties				
Moisture Absorption		0.02	%	IPC-650 2.6.2.1
Dielectric Breakdown		45	KV	IPC-650 2.5.6 (ASTM D 149)
Flammability Rating		V-0		UL-94

* As reported by NASA. See http://outgassing.nasa.gov/og_disclaimer.html

Typical Thickness			
Inches	**mm**	**Inches**	**mm**
0.0050	0.13	0.0300	0.76
0.0100	0.25	0.0600	1.52
0.0200	0.51	0.1100	2.79

Available Copper Cladding						
Designation	**Weight**	**Copper Thickness**		**Rz ISO Treated Side**		**Description**
ULPH	½ oz./sq. ft.	~ .0007"	~ 18 µm	67µin	1.7 µm	Ultra low profile / Electrodeposited
ULP1	1 oz./sq. ft.	~ .0014"	~ 35 µm	59µin	1.5 µm	Ultra low profile / Electrodeposited
CH	½ oz./sq. ft.	~ .0007"	~ 18 µm	236µin	6 µm	Very low profile / Electrodeposited
C1	1 oz./sq. ft.	~ .0014"	~ 35 µm	236µin	6 µm	Very low profile / Electrodeposited
C2	2 oz./sq. ft.	~ .0028"	~ 70 µm	433µin	11 µm	Very low profile / Electrodeposited
CLH	½ oz./sq. ft.	~ .0007"	~ 18 µm	200µin	5.1 µm	Reverse Treated / Electrodeposited
CL1	1 oz./sq. ft.	~ .0014"	~ 35 µm	200µin	5.1 µm	Reverse Treated / Electrodeposited

Available Sheet Sizes			
Inches	**mm**	**Inches**	**mm**
12 x 18	305 x 457	24 x 36	610 x 914
16 x 18	406 x 457	18 x 48	457 x 1,220
18 x 24	457 x 610	36 x 48	914 x 1,220
16 x 36	406 x 914		

* All test data provided are typical values and not intended to be specification values. For review of critical specification tolerances, please contact a company representative directly.
* Standard panel size is 18” x 24” (457 mm x 610 mm).
* Minimum available thickness is from 0.0025” (0.063 mm).
* Please contact AGC for availability of additional thicknesses, other sizes & any other type of cladding.

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